UTKARSH SHARMA

Software Engineer & Data Science Undergraduate

B.Tech in Computer Science & Engineering (Specialization in Data Science)

GitHub: github.com/shutkarshX

LinkedIn: linkedin.com/in/utkarsh-sharma

Live Application: tein.onrender.com

2,600+	1,830+	190+	Production
ALL-TIME COMMITS	CONTRIBUTIONS	AUTOMATED UNIT TESTS	LIVE CLOUD DEPLOYS

EXECUTIVE PROFILE

Hands-on software engineer and Computer Science undergraduate specializing in Data Science. Demonstrated track record of designing, building, and deploying real-world software systems — spanning production full-stack web platforms on Render, native desktop system applications in Rust, AI-driven crisis command centers, and browser automation extensions. Deep commitment to software quality, automated testing rigor (over 190+ passing unit and E2E tests), and clean system architecture.

TECHNICAL COMPETENCIES

Programming Languages

`Python` `TypeScript` `JavaScript` `Rust` `C (C99)` `Java` `SQL` `R`

Frameworks & Libraries

`FastAPI` `Flask` `React` `Vite` `Tailwind CSS` `Tauri 2` `Pandas` `Scikit-Learn`

Systems & Architecture

`PWA (Service Workers)` `REST APIs` `Docker` `Win32 APIs` `Asyncio` `TCP Sockets` `Leaflet GIS`

DevOps, Testing & Tools

`Git / GitHub Actions` `Render Cloud` `Pytest (Automated)` `Chrome MV3` `Linux / Bash` `Gunicorn`

VERIFIED ENGINEERING PROJECTS

TEIN (Bunku-Bhaiya) — Attendance Intelligence & Campus Hub • LIVE PRODUCTION

Live App ↗ Showcase Repo ↗

A full-stack, mobile-first university platform deployed live on Render to solve attendance tracking and campus activity discovery across the semester cycle.

- **Bunk & Recovery Calculator:** Implemented decision support algorithms calculating exact safe lecture skip thresholds and mandatory recovery lecture chains to maintain the institutional 75% requirement.
- **What-If Scenario Simulator:** Engineered a forward-looking planner allowing students to simulate hypothetical future leaves or attendance streaks before taking time off.
- **Campus Community Hub:** Built a centralized discovery platform indexing profiles, domains, and event announcements for 20+ university clubs and societies.
- **Mobile-First PWA:** Architected with service worker offline shell caching (`/sw.js`) enabling reliable operation over weak campus networks.

`Python 3` `Flask` `Docker` `PWA` `Gunicorn` `Render Cloud` `25 Unit Tests`

RESQ-AI — Intelligent Disaster Response Command Center

GitHub Repo ↗

Emergency response operations dashboard designed for municipal crisis management, real-time rescue prioritization, and geospatial resource routing during flood disasters.

- **AI Incident Triage:** Integrated Google Gemini AI to analyze raw citizen emergency dispatches, extract geo-coordinates, and dynamically classify incident severity levels.
- **Geospatial Command Map:** Rendered interactive Leaflet maps with live incident markers, rescue convoy tracking, and shelter capacity overlays.
- **High-Throughput API:** Built with FastAPI using async routing and typed Pydantic models with OpenAPI/Swagger interactive testing endpoints.

`FastAPI` `React` `Gemini AI` `Leaflet Maps` `Swagger` `Python Async`

Layer-Flow — Transparent Screen Drawing & Desktop Canvas

GitHub Repo ↗

A native, lightweight Windows desktop application allowing presenters and educators to draw, highlight, and annotate directly over any active application in real time.

- **Native Rust & Tauri 2 Architecture:** Sub-50MB memory footprint and zero UI latency, avoiding the heavy overhead of standard Chromium-based wrappers.
- **Win32 Window Management:** Leveraged low-level OS hooks for global keyboard hotkeys and click-through pass-through state toggling.

`Rust` `Tauri 2` `React` `Win32 API` `Systems Programming`

Speedo — Coursework Progression & Adaptive Video Controller

GitHub Repo ↗

A Chrome Extension (Manifest V3) designed to automate learning progression, bypass arbitrary player speed restrictions, and safeguard coursework progression.

- **CSP-Safe Injection:** Implemented a Main World execution bridge complying with strict Manifest V3 Content Security Policies without triggering security violations.
- **Modal Dismissal Guard:** Automatically intercepts and handles idle detection popups to maintain continuous coursework playback.

`JavaScript` `Chrome Extension MV3` `DOM Automation` `CSP Safety`

Integrated National Citizen Services Platform & CPU Scheduling Lab

GitHub Repo ↗

Academic Operating Systems lab suite implementing and validating fundamental process scheduling algorithms with an exhaustive automated test suite.

- **Core Scheduling Implementations:** Built First-Come-First-Served (FCFS), Shortest Job First (SJF Preemptive & Non-Preemptive), Priority Scheduling, and Round Robin.
- **166 Automated Tests:** Comprehensive test suite covering turnaround time calculations, process starvation edge cases, context-switch tracking, and CPU utilization metrics.

`Python` `Operating Systems` `CPU Scheduling` `166 Automated Tests`

Data Science & Predictive Modeling Suite (Prodigy Infotech)

GitHub Repo ↗

End-to-end data science projects covering exploratory data analysis, machine learning classification, sentiment analysis, and geospatial accident clustering.

- **Predictive Decision Tree Modeling:** Trained and hyperparameter-tuned decision tree classifiers on customer demographics to predict subscription behavior.
- **EDA & Sentiment Mining:** Processed Titanic survival factors and extracted sentiment patterns from social media feeds using NLP pipelines.

`Python` `Pandas` `Scikit-Learn` `Seaborn` `Exploratory Data Analysis`